

Sciex ZT Scanning-DIA: Change Audit & Next Steps

What we changed, keeping it minimal, and where to go

feat/sciex-zt-scanning

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1 Where we are

The full pipeline now runs end-to-end on ZT scanning-DIA data. A 3-file run (MBR on) completes in ~ 7.3 min and yields ~ 85.7 k precursor rows / ~ 13.2 k protein-group rows, with clean per-cycle chromatograms and sane peak areas. Everything ZT is gated on the environment variable `PIONEER_ZT_METASCAN_K` (k); with $k = 0$ the code is byte-identical to the base branch, so **no other dataset is affected today**.

2 What we have changed for ZT (audit)

Net footprint vs. the branch base: **9 files, +523 / -22 lines**. Grouped by role:

File	Δ lines	Role
<i>Core acquisition model (the actual ZT algorithm)</i>		
LibrarySearch.jl	+173	two-width square quad (frag-index center bin ≈ 1 m/z; deconv $(2k+1)$ S box), meta-scan candidate expansion, center-bin filter, <code>_zt_bin_step</code>
MainSearch.jl	+32	call the collapse before the per-file LGBM
metascan_collapse.jl	+132 (new)	collapse $\pm k$ bins $\rightarrow$ 1 meta-PSM + 13 weight cols + 9 profile features
IntegrateChromatogramsSearch.jl	+46	wide extraction quad + per-cycle collapse wiring
...IntegrateChroms/utils.jl	+88	<code>collapse_chromatograms_to_metascans</code> (triangle-weighted)
importScripts.jl	+1	include the new file
<i>Tuning / calibration (we want this minimal)</i>		
ParameterTuningSearch.jl	+10	env-gated shuffle! of calibration scan sampling
fit_intensity_mass_error.jl	+46	free (non-monotone) bias spline, dispatched ZT $\rightarrow$ free else monotone
<i>Orthogonal (a pre-existing bug, not ZT)</i>		
score_psms.jl	+16	skip MBR when only one raw file (committed separately)

2.1 Parameter-tuning minimality — assessment

This is the part we most want to keep small, and it already is:

- **Calibration shuffle (+10 lines).** Entirely env-gated on `PIONEER_ZT_METASCAN_K > 0`. Non-ZT runs never execute it $\Rightarrow$ byte-identical. *Keep*.
- **Free bias spline (+46 lines).** Adds a non-monotone spline option and dispatches ZT $\rightarrow$ free, else the original monotone-best fit, so non-ZT results are unchanged. Its ZT benefit was never isolated in an A/B. *Candidate to drop* if it isn't clearly helping — that would take Parameter Tuning down to the single 10-line shuffle.
- **Already reverted (dead ends):** the ± 3 m/z tuning window (**worse:** 20,318 $\rightarrow$ 18,783) and the PSM-target 1200 $\rightarrow$ 5000 (**neutral**). Neither is in the tree.

Bottom line: the only tuning change we truly need is the 10-line, env-gated calibration shuffle. Everything else ZT lives in the acquisition-model code, not in the tuning/calibration logic.

3 Next steps

3.1 1. Efficiency — the $13\times$ candidate blow-up (highest value)

Main Search is **56% of runtime** because meta-scan expansion multiplies candidates by $2k+1$ ($\approx 13\times$ at $k=6$), producing ~ 29 M deconv PSMs per file; the wide-quad chromatogram extraction has the same blow-up. Levers:

- **Option B chromatogram collapse:** accumulate $\sum w_j t_j$ per (precursor, cycle) *inside* `build_chromatograms` and emit one point per cycle, instead of materialising $13\times$ points then collapsing ($13\times$ fewer objects + memory).
- **Prune profile features $9 \rightarrow \sim 3$** (`zt_tri_cosine`, `zt_tail_frac`, `zt_entropy`); drop the 13 stored weight columns once features are computed.
- **Bound the expansion** to bins that actually carry signal, and confirm `expand_to_metascans!` never over-expands beyond center-matched precursors.

3.2 2. Auto-detect the acquisition (remove the env flag)

Replace `PIONEER_ZT_METASCAN_K` with values *read from the raw file*: the per-scan center-m/z step S (does the center advance within a cycle?) and the physical window width W . This makes ZT handling automatic and removes a manual knob.

3.3 3. Correctness / quant

DIA-NN reference on these exact three files (the only fair yardstick vs. the paper); entrapment / EFDR; verify quant CV against the reported $\sim 2.8\%$.

4 The unification: treat *all* data as ZT

The cleanest long-term design (and the stated goal): parameterise the search by two per-acquisition numbers and run *one* algorithm.

Parameter	Conventional DIA	ZT scanning
S = center-m/z step per scan	0	≈ 1.022 m/z
W = physical window width $\Rightarrow k$ (bins each side)	reported isolation width 0	swept window = $(k+1)S$ ≈ 6

With k derived from (S, W) , every ZT stage degenerates *exactly* to today’s conventional path when $S = 0 \Rightarrow k = 0$:

ZT stage	Behaviour at $k = 0$ (conventional)
meta-scan expansion	no neighbours $\Rightarrow$ no-op
two-width quad (deconv box $(2k+1)S$)	box = reported isolation width $\Rightarrow$ unchanged
center-bin filter	the window <i>is</i> the bin $\Rightarrow$ identity
chromatogram collapse	one bin per cycle already $\Rightarrow$ identity

So the ZT code path **becomes the general path**, and conventional DIA is just the special case $S = 0$. Payoff: delete the `if _zt` branches and the env gate, one code path to maintain, and ZT is continuously exercised by the existing regression suite (conventional data flowing through the same kernels).

What it takes.

1. Derive $(S, W) \rightarrow k$ from the raw file (step 2 above).
2. Replace each `_zt/_zt_main` gate with the k -parameterised form, ensuring the $k = 0$ reduction is *provably* identical to the current conventional code.
3. **Byte-identical regression tests:** run existing conventional datasets through the unified path and require identical output to today’s ($k = 0$ must not move a single number).
4. Keep the two *non*-acquisition tuning tweaks (calibration shuffle, bias spline) decided separately — they are sampling/fitting choices, not part of the (S, W) model, and should stay gated on $k > 0$ (or be dropped) rather than folded into the unified kernel.

Risk. The only real risk is the $k = 0$ byte-identical guarantee: the two-width quad, the wide extraction quad, and the collapse must each reduce *to the exact current arithmetic*, not merely “close”. The regression tests in step 3 are the gate that makes this safe.